INTERNSHIP PROJECT REPORT

AWS Networking:

VPC Peering with a Bastion Host

Two private VPCs, one public entry point, verified with ping and SSH

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AWS Region used	Asia Pacific (Mumbai), ap-south-1
Project code	EWB: VPC Peering with Bastion Host
Submission date	21/10/2026

1. Introduction and Architecture

Objective. To build two isolated Amazon VPCs, and connect them with a VPC peering connection, and then reach the private servers in both VPCs through a single public bastion host, then prove the design works with ping and SSH tests.

Scope. Region ap-south-1 (Mumbai). Three t3.micro Amazon Linux 2023 instances: one bastion in the public subnet of VPC A, one private server in VPC A, and one private server in VPC B. Only the bastion has a public IP address. VPC B has no internet gateway.

1.1 Address plan

Resource	Name	CIDR block	VPC
VPC	ewb-vpc-a	10.10.0.0/16	-
Public subnet	ewb-a-public	10.10.1.0/24	ewb-vpc-a
Private subnet	ewb-a-private	10.10.2.0/24	ewb-vpc-a
VPC	ewb-vpc-b	10.20.0.0/16	-
Private subnet	ewb-b-private	10.20.1.0/24	ewb-vpc-b

The two 1.2 Architecture diagram

VPC ranges do not overlap, which VPC peering requires.

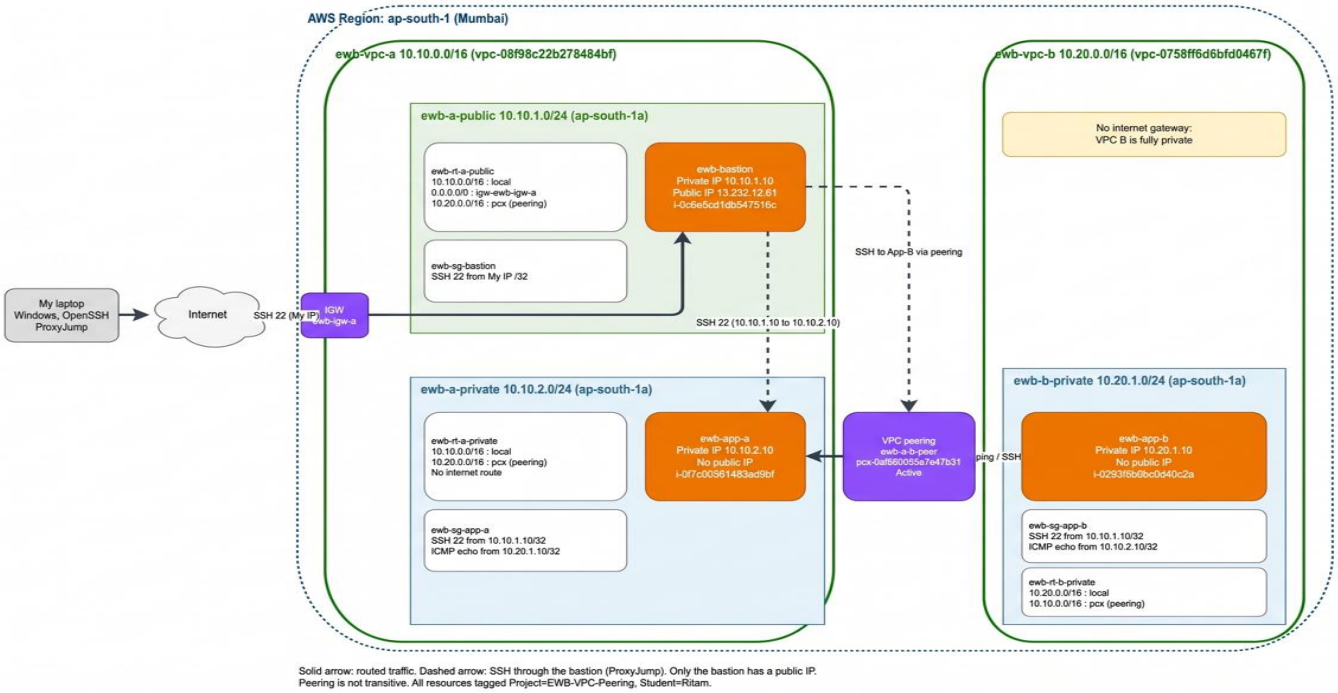


Figure 1: Architecture of the VPC peering project (drawn in draw.io)

2. Resource Inventory

All resources were created in Asia Pacific (Mumbai), ap-south-1, and tagged Project = EWB-VPC-Peering.

Resource	Name	ID	Details
VPC	ewb-vpc-a	vpc-08f98c22b278484bf	10.10.0.0/16
Subnet	ewb-a-public	subnet-00773dc0eaa19ad14	10.10.1.0/24, ap-south-1a
Subnet	ewb-a-private	subnet-09d795929ac18f6e0	10.10.2.0/24, ap-south-1a
Internet gateway	ewb-igw-a	igw-0cb1d4b718da37e06	Attached to ewb-vpc-a
Route table	ewb-rt-a-public	rtb-02941e49cba7baf0e	Associated: ewb-a-public
Route table	ewb-rt-a-private	rtb-04cf5c58ec38c31ab	Associated: ewb-a-private
VPC	ewb-vpc-b	vpc-0758ff6d6bfd0467f	10.20.0.0/16
Subnet	ewb-b-private	subnet-0e87c45434a689358	10.20.1.0/24, ap-south-1a
Route table	ewb-rt-b-private	rtb-04cfb0052b71b1a5a	Associated: ewb-b-private
Peering	ewb-a-b-peer	pcx-0af280859a7e47b31	Active, vpc-a to vpc-b
Security group	ewb-sg-bastion	sg-0aa8423581aeedaed	ewb-vpc-a
Security group	ewb-sg-app-a	sg-063d5b02401930306	ewb-vpc-a
Security group	ewb-sg-app-b	sg-03e5d27d825d740af	ewb-vpc-b
Instance	ewb-bastion	i-0c6e5cd1db547516c	10.10.1.10, public 13.232.12.61
Instance	ewb-app-a	i-0f7c08561483ad9bf	10.10.2.10, no public IP
Instance	ewb-app-b	i-0293fbb66c0d40c2a	10.20.1.10, no public IP
Key pair	ewb-lab	RSA, .pem	Used for all three instances

3. Network Build: VPCs, Subnets and Internet Gateway

VPC A (ewb-vpc-a, 10.10.0.0/16) holds the bastion and App-A. VPC B (ewb-vpc-b, 10.20.0.0/16) holds App-B. Each VPC has its own subnet range, and all three subnets are in ap-south-1a.

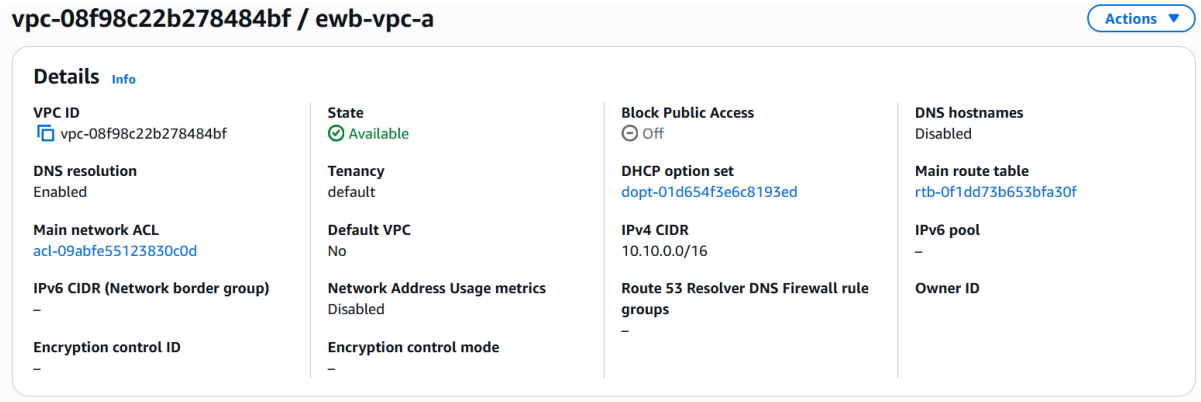


Figure 2: VPC A (ewb-vpc-a) created with CIDR 10.10.0.0/16

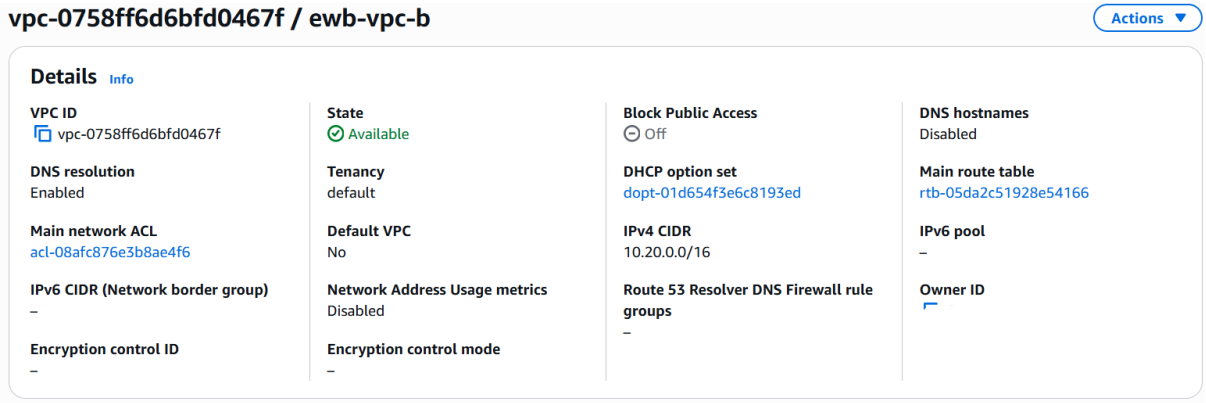


Figure 3: VPC B (ewb-vpc-b) created with CIDR 10.20.0.0/16

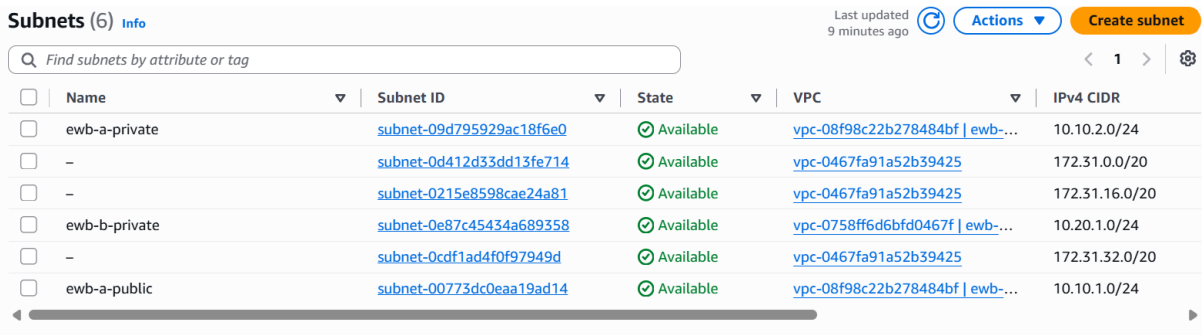


Figure 4: Subnets ewb-a-public, ewb-a-private and ewb-b-private, all Available

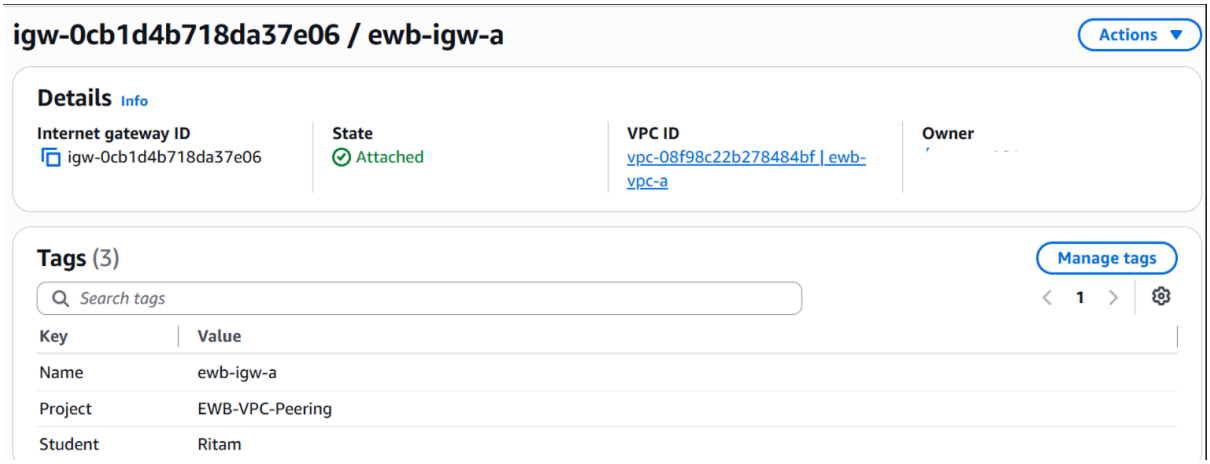


Figure 5: Internet gateway ewb-igw-a attached to ewb-vpc-a

4. Route Tables Before Peering (Checkpoint 01)

A route table decides where each subnet's traffic goes. Every subnet is associated explicitly with its own table. The public table has a default route to the internet gateway. The private tables have only the local route, so App-A and App-B cannot reach the internet.

4.1 ewb-rt-a-public

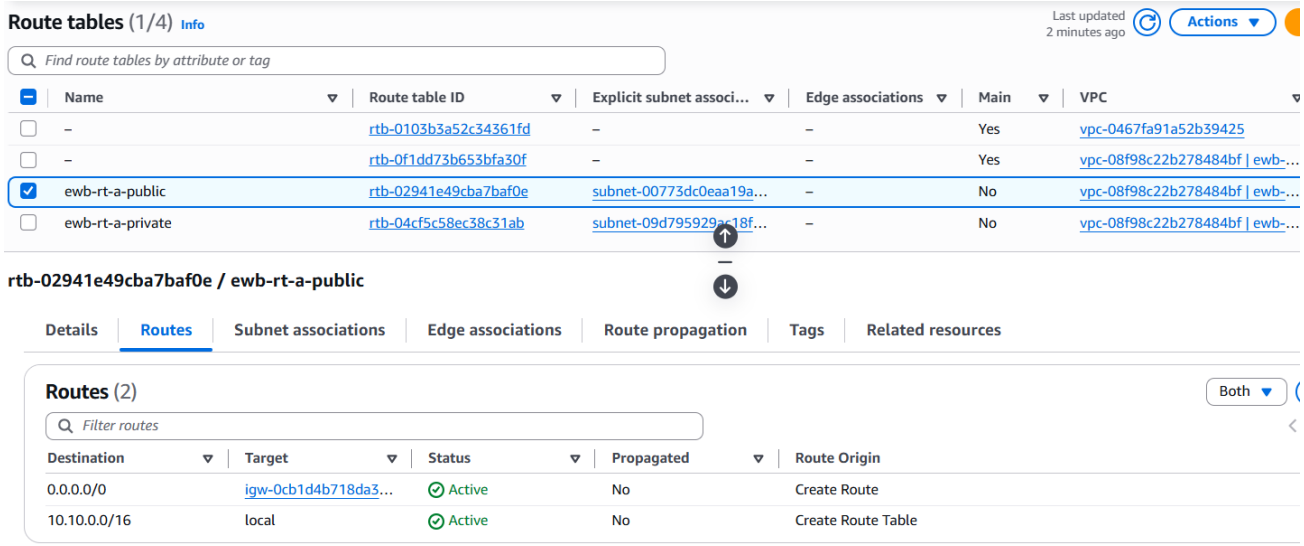


Figure 6: ewb-rt-a-public routes before peering (internet route present)

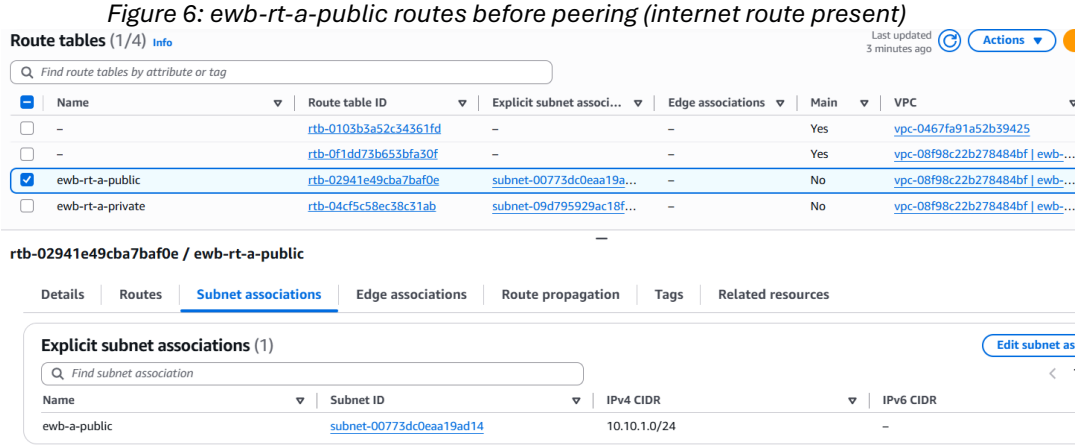


Figure 7: ewb-rt-a-public associated with ewb-a-public

4.2 ewb-rt-a-private

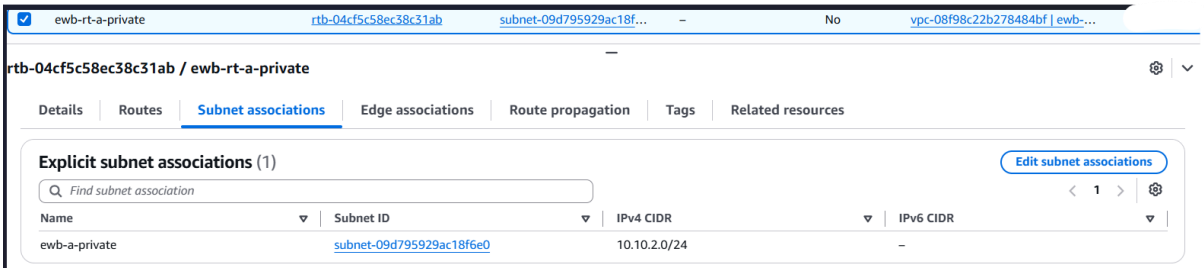


Figure 8: ewb-rt-a-private routes before peering (local route only)

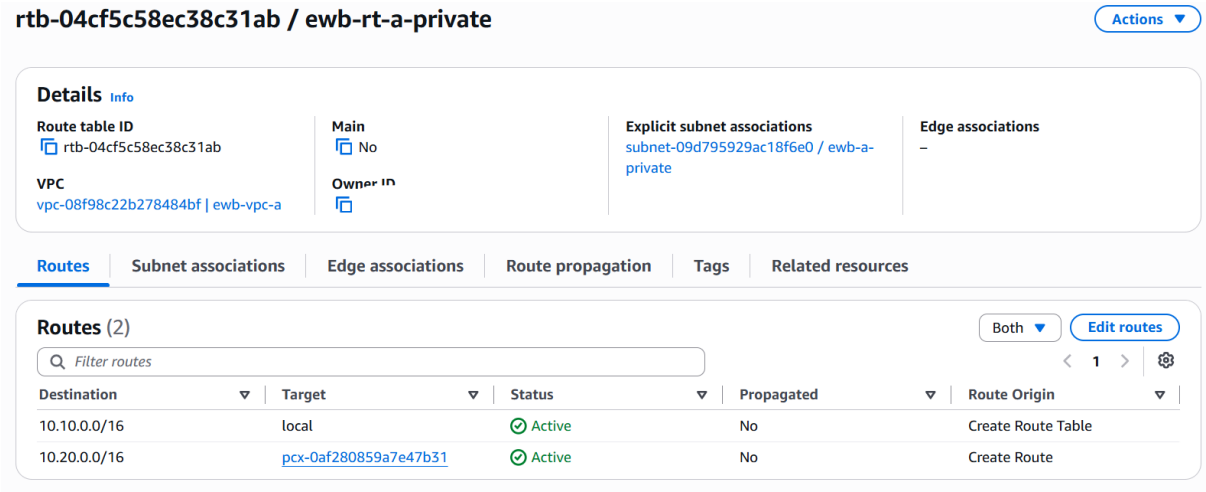


Figure 9: ewb-rt-a-private associated with ewb-a-private

4.3 ewb-rt-b-private

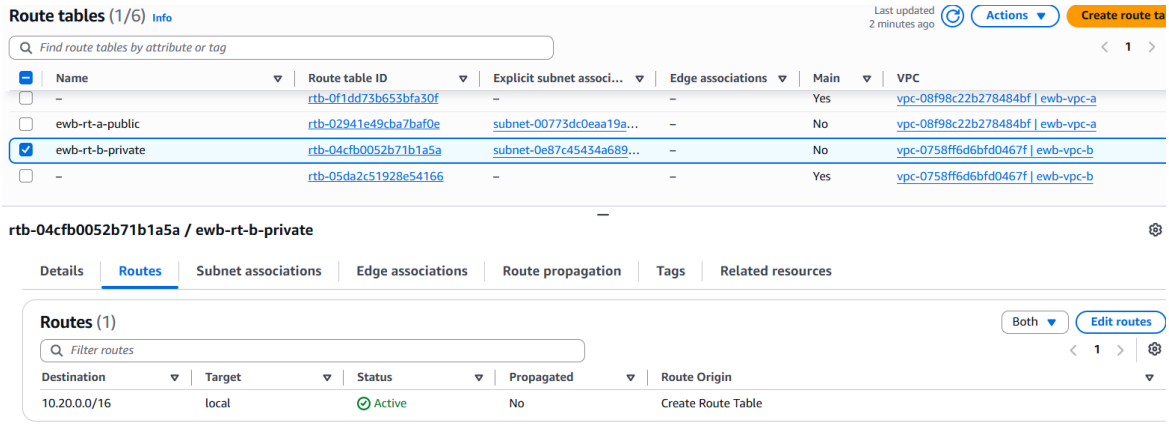


Figure 10: ewb-rt-b-private routes before peering (local route only)

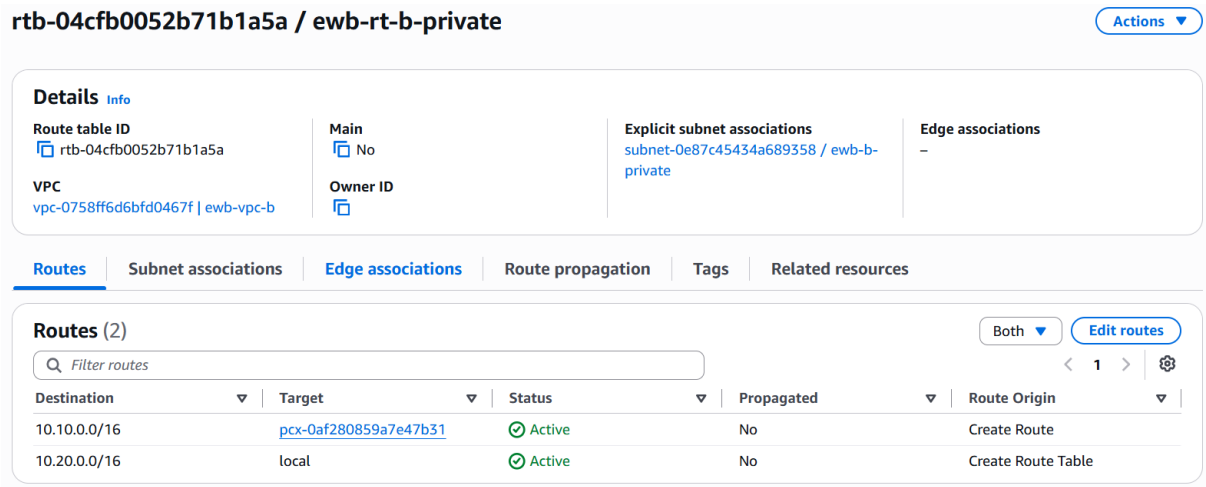


Figure 11: ewb-rt-b-private associated with ewb-b-private

5. Security Groups

Each server has its own security group that allows only what the lab needs. Outbound rules were left at the default. SSH is never open to the whole internet in the final design.

Group	VPC	Inbound rule	Source	Purpose
ewb-sg-bastion	ewb-vpc-a	SSH 22	My IP /32	Admin entry point
ewb-sg-app-a	ewb-vpc-a	SSH 22	10.10.1.10/32	From the bastion only
ewb-sg-app-a	ewb-vpc-a	ICMP echo request	10.20.1.10/32	Ping from App-B
ewb-sg-app-b	ewb-vpc-b	SSH 22	10.10.1.10/32	From the bastion only
ewb-sg-app-b	ewb-vpc-b	ICMP echo request	10.10.2.10/32	Ping from App-A

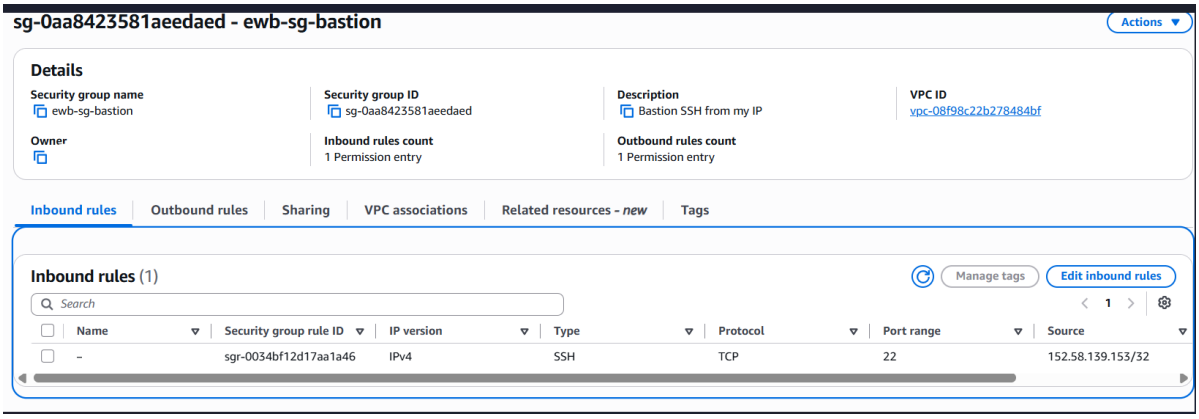


Figure 12: Inbound rules of ewb-sg-bastion

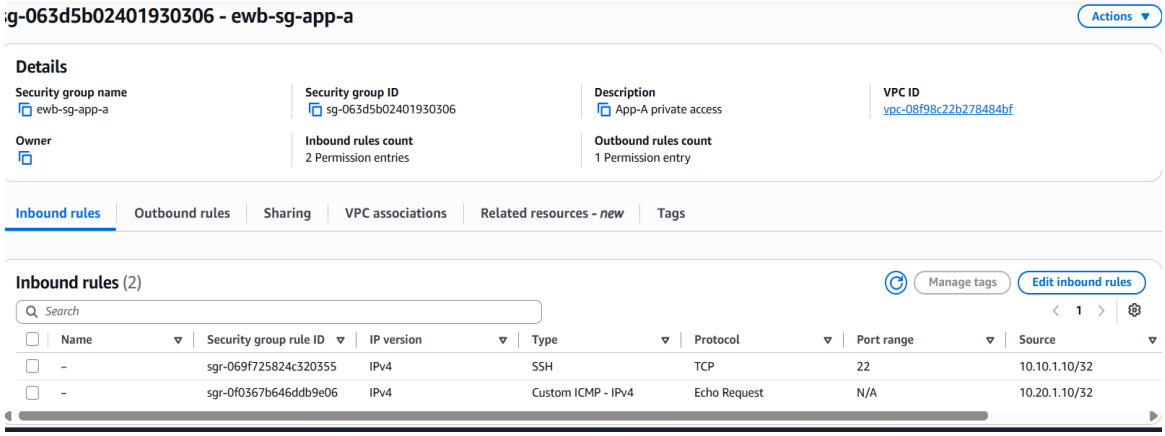


Figure 13: Inbound rules of ewb-sg-app-a

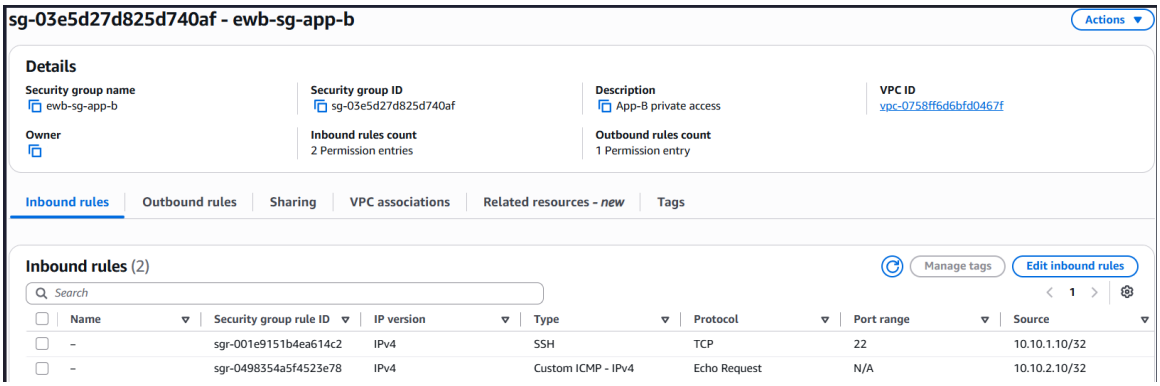


Figure 14: Inbound rules of ewb-sg-app-b

6. EC2 Instances and SSH Setup (Checkpoint 02)

Three t3.micro instances running Amazon Linux 2023 were launched with the key pair ewb-lab. Each has a fixed private IP. Only the bastion has a public IP. The private key stays on my laptop and is not included in this report.

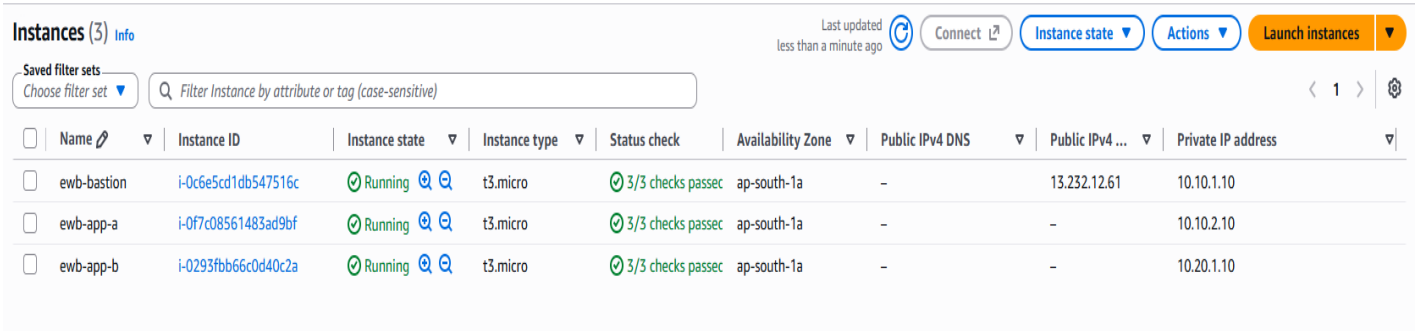


Figure 15: Three instances running; only ewb-bastion has a public IP

6.1 SSH configuration with ProxyJump

My laptop connects to the bastion, which forwards the connection to the private servers. The configuration file below contains only host names and IP addresses, no key material.

```
Host ewb-bastion
  Hostname 13.232.12.61
  User ec2-user
  IdentityFile ~/.ssh/ewb-lab.pem
  IdentitiesOnly yes

Host ewb-app-a
  Hostname 10.10.2.10
  User ec2-user
  IdentityFile ~/.ssh/ewb-lab.pem
  ProxyJump ewb-bastion

Host ewb-app-b
  Hostname 10.20.1.10
  User ec2-user
  IdentityFile ~/.ssh/ewb-lab.pem
  ProxyJump ewb-bastion
```

7. Baseline Tests Before Peering

Before creating the peering connection I confirmed SSH access and showed that the two VPCs could not reach each other.

Test	From to	Command	Expected	Actual
A	Laptop to bastion	ssh ewb-bastion	Login works	Login works, IP 10.10.1.10
B	Laptop to App-A	ssh ewb-app-a	Login via bastion	Login works, IP 10.10.2.10
1	App-A to App-B	ping -c 4 10.20.1.10	No replies	100% packet loss

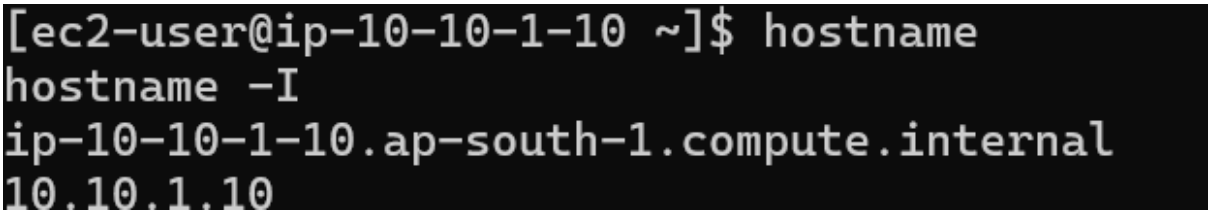


Figure 16: Bastion: SSH from my laptop works, private IP 10.10.1.10

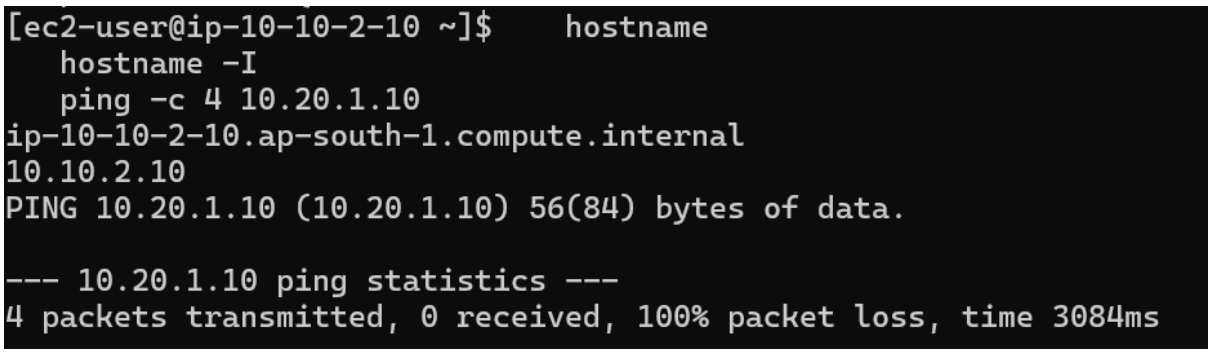


Figure 17: App-A: baseline ping to App-B before peering, 100% packet loss

Why it failed. The two VPCs are separate networks. Without a peering connection and routes, App-A has no path to 10.20.1.10.

8. VPC Peering and Routes (Checkpoints 03 and 04)

I created the peering connection ewb-a-b-peer with ewb-vpc-a as requester and ewb-vpc-b as accepter in the same account and Region, and accepted the request. Peering alone moves no traffic, so I added a route to the peering connection in each of the three route tables.

Route table	Destination	Target
ewb-rt-a-public	10.20.0.0/16	pcx-0af280859a7e47b31
ewb-rt-a-private	10.20.0.0/16	pcx-0af280859a7e47b31
ewb-rt-b-private	10.10.0. 0/16	pcx-0af280859a7e47b31

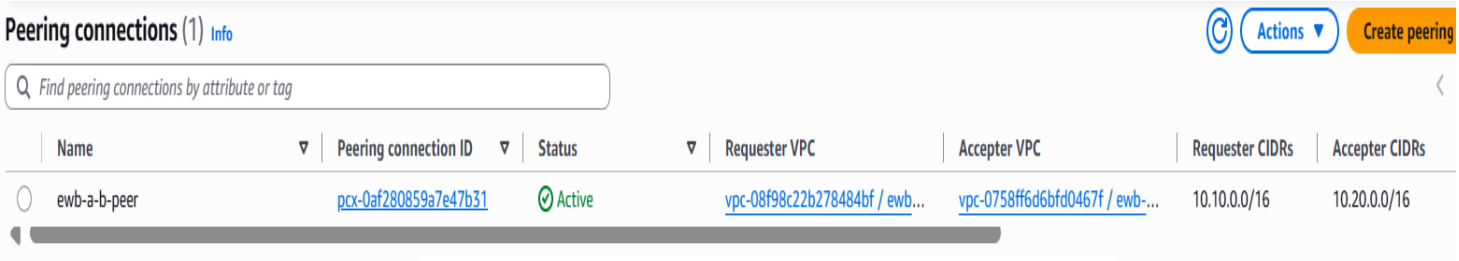


Figure 18: Peering connection ewb-a-b-peer is Active

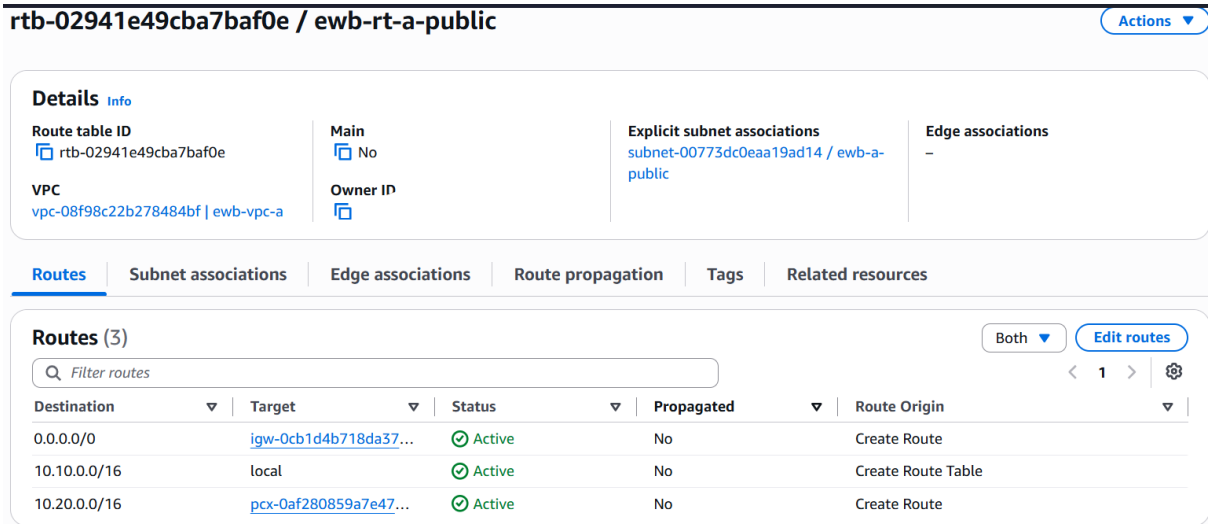


Figure 19: ewb-rt-a-public after peering

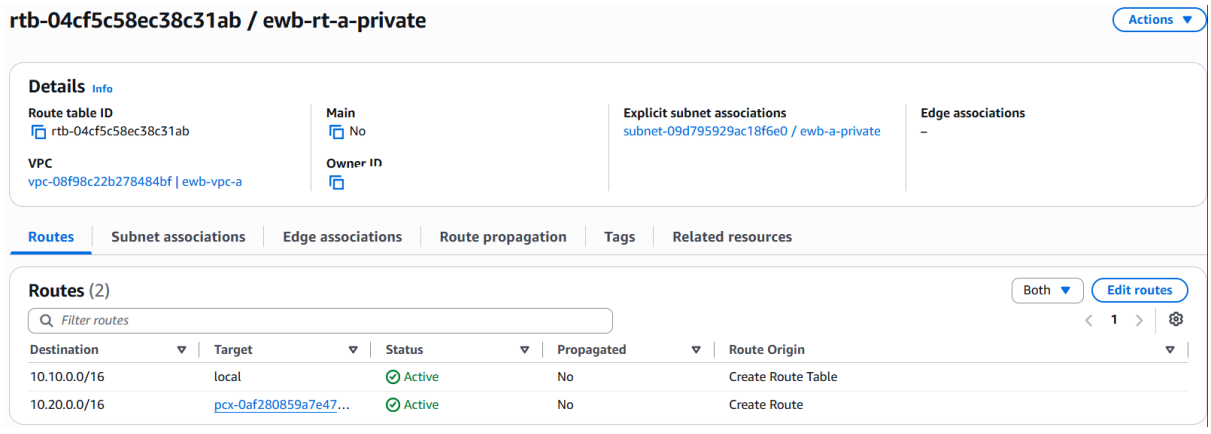


Figure 20: ewb-rt-a-private after peering

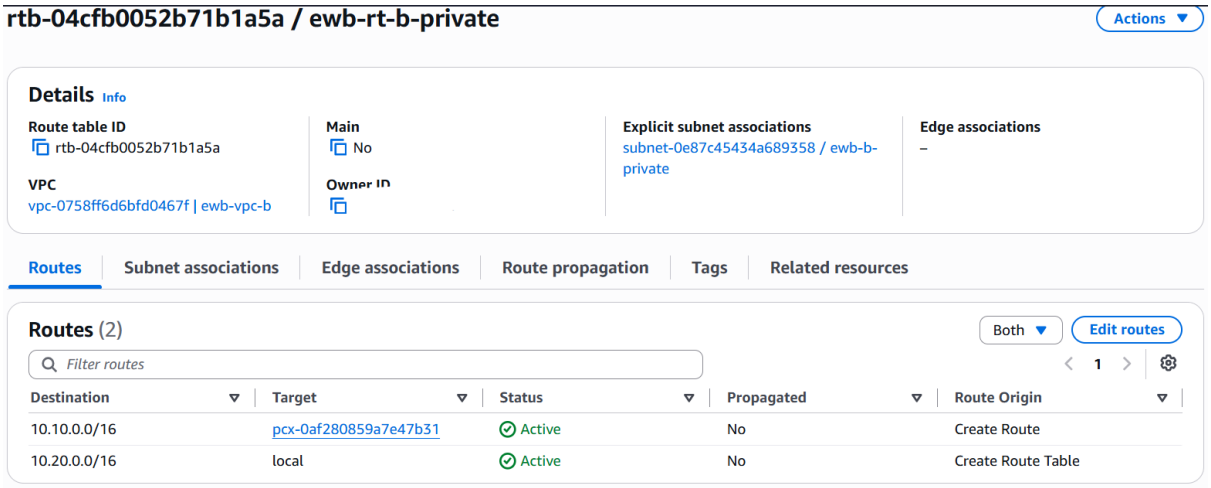


Figure 21: ewb-rt-b-private after peering

8.1 Forward path and return path

Forward path. A packet from App-A (10.10.2.10) to App-B (10.20.1.10) matches the route 10.20.0.0/16 in ewb-rt-a-private, goes through the peering connection, and arrives in ewb-vpc-b.

Return path. App-B's reply to 10.10.2.10 matches the route 10.10.0.0/16 in ewb-rt-b-private and goes back through the same peering connection. Without this route the replies would be dropped, and the ping would fail.

Bastion path. The bastion sits in the public subnet, so ewb-rt-a-public also needs the 10.20.0.0/16 route. That is what lets the bastion open SSH sessions to App-B.

Not transitive. Peering connects only the two VPCs. VPC B cannot use VPC A's internet gateway through the peering connection.

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Return path. App-B's reply to 10.10.2.10 matches the route 10.10.0.0/16 in ewb-rt-b-private and goes back through the same peering connection. Without this route the replies would be dropped, and the ping would fail.

Bastion path. The bastion sits in the public subnet, so ewb-rt-a-public also needs the 10.20.0.0/16 route. That is what lets the bastion open SSH sessions to App-B.

Not transitive. Peering connects only the two VPCs. VPC B cannot use VPC A's internet gateway through the peering connection.

9. Verification Tests After Peering

Test	Description	Command	Expected	Actual / Figure
1	Baseline, before peering	ping -c 4 10.20.1.10	100% loss	100% loss (Figure 17)
2	App-A to App-B	ping -c 4 10.20.1.10	0% loss	4/4 replies, 0% loss (Figure 22)
3	App-B to App-A	ping -c 4 10.10.2.10	0% loss	4/4 replies, 0% loss (Figure 23)
4	Private SSH to App-B	ssh ewb-app-b	Login works	Login works (Figure 23)
5	No public IP on app servers	Console check	Empty field	Confirmed (Figure 24)

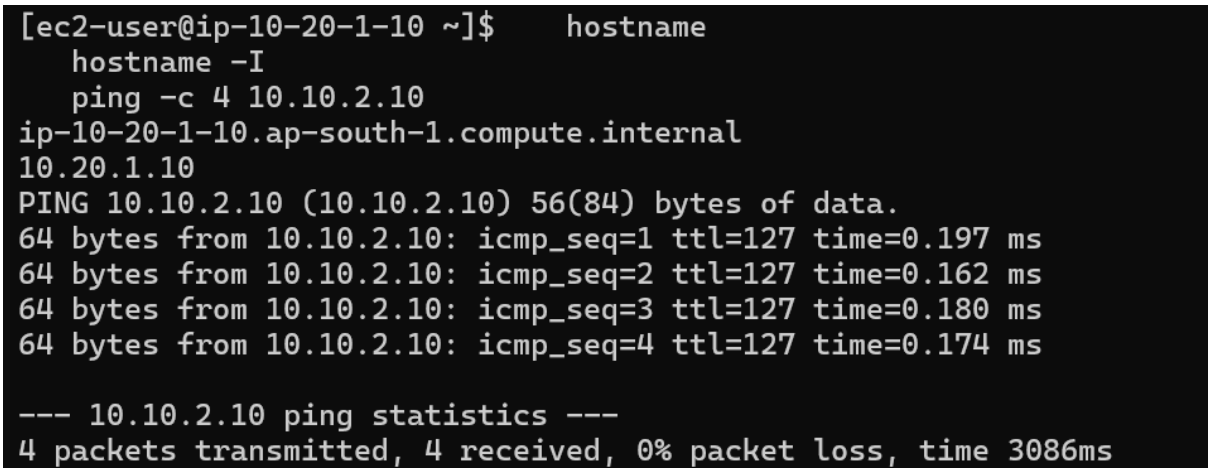


Figure 22: App-A to App-B after peering, 4/4 replies, 0% packet loss

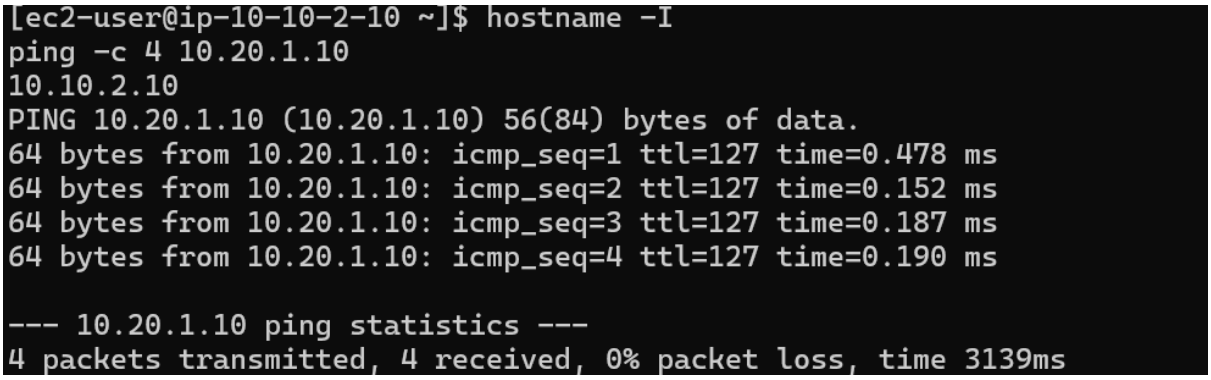


Figure 23: App-B: private SSH works and ping to

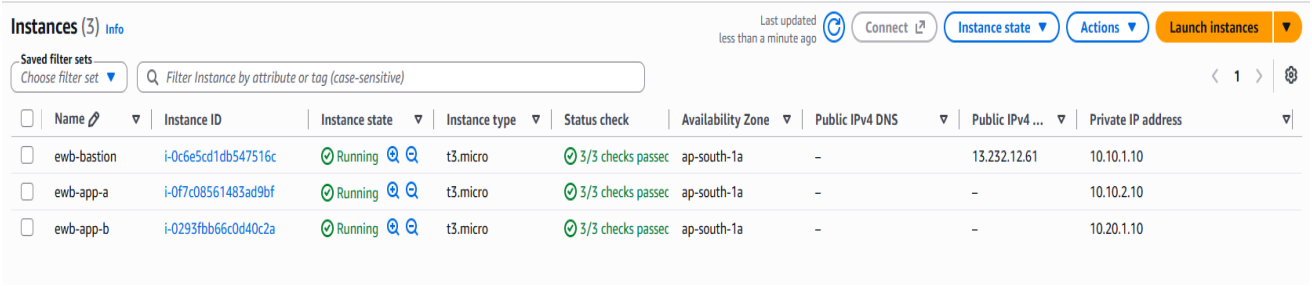


Figure 24: App-A and App-B have no public IP address

10. Controlled Failure and Recovery

To show that security groups control traffic in addition to routes, I removed only the ICMP rule (source 10.10.2.10/32) from ewb-sg-app-b and left the SSH rule in place. The ping from App-A to App-B then failed. I added back the identical rule and the ping recovered.

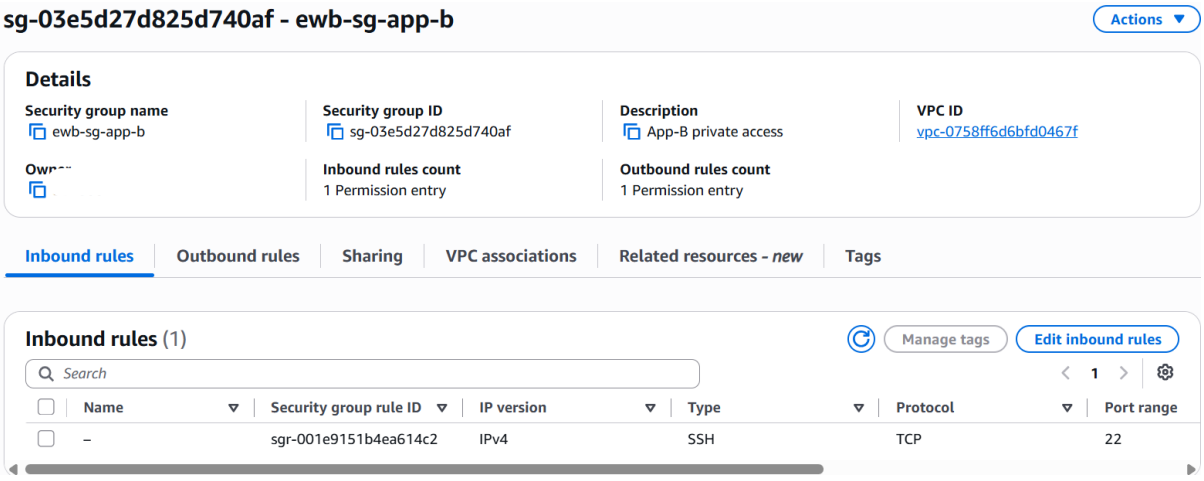


Figure 25: ICMP rule removed from ewb-sg-app-b

```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3126ms
```

Figure 26: App-A to App-B fails after the ICMP rule is removed

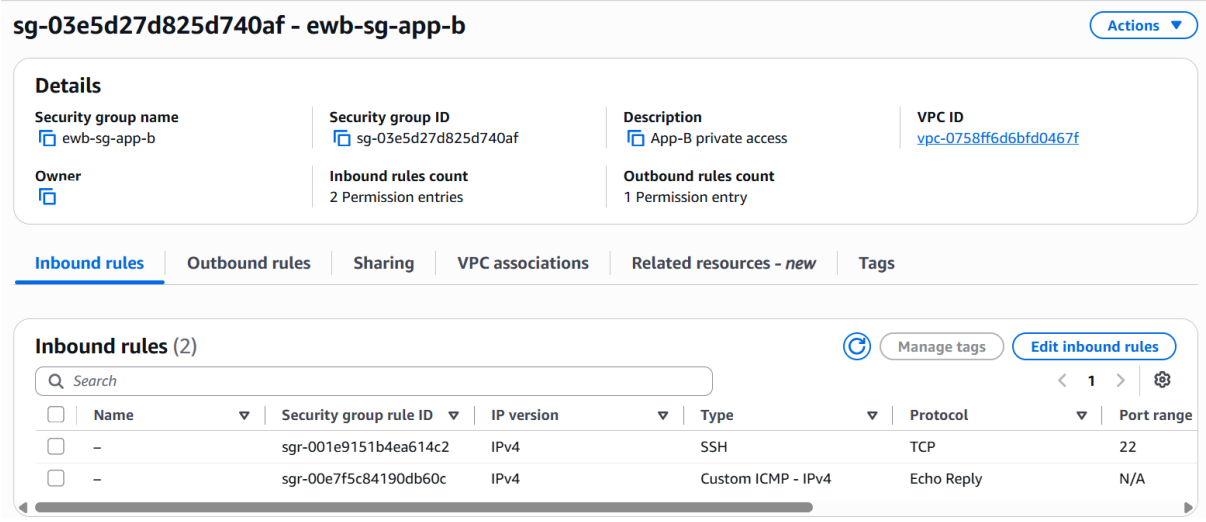


Figure 27: ICMP rule restored in ewb-sg-app-b

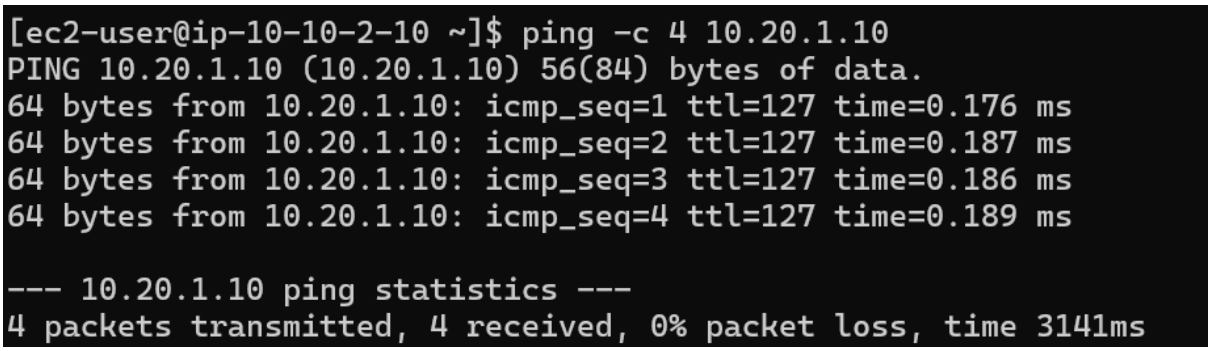


Figure 28: Ping recovers after the rule is restored

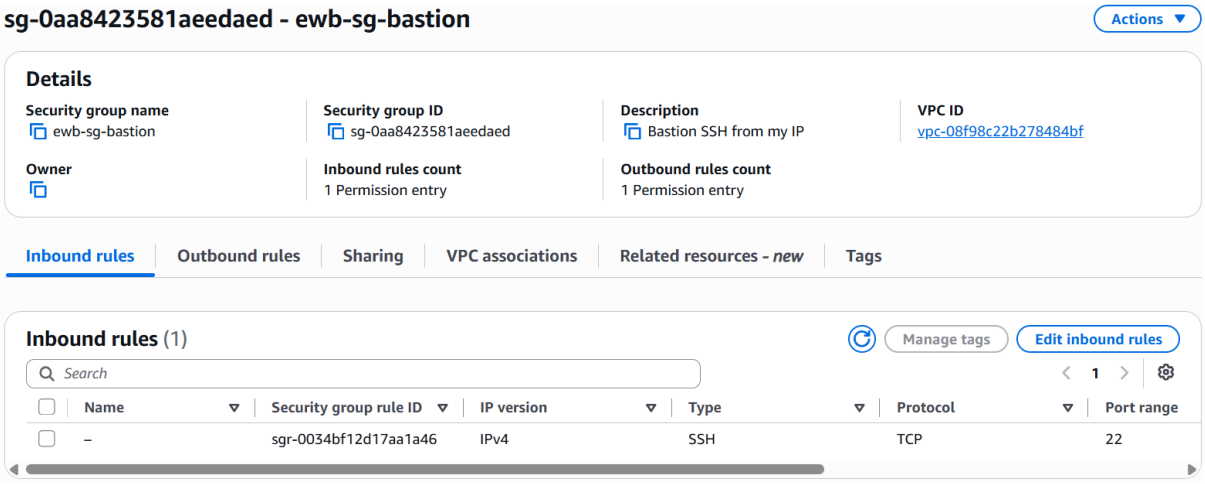
What this shows. Correct routes are necessary but not enough: the destination's security group must also allow the traffic.

11. Security Review and Issues Faced

11.1 Bastion access from a mobile hotspot

I worked on a mobile hotspot. SSH to the bastion timed out even though the security group allowed my IP address as shown by an IP-check website. Widening the rule to my provider's /24 did not help, but briefly allowing SSH from anywhere did. This most likely means my carrier uses carrier-grade NAT, so the address AWS saw was different from the one I measured. I used the open rule only for a short test window.

After the tests, I restricted the SSH rule again to my own IP address (Figure 29).



11.2 Other issues and fixes

Issue	Cause	Fix
VPC A first showed 10.0.0.0/16	The console pre-fills a default CIDR	Deleted it and recreated it with 10.10.0.0/16
ssh: could not resolve hostname	The SSH config file was empty (0 bytes)	Rewrote the file with PowerShell and re-tested
SSH to the bastion timed out	Source IP did not match on the hotspot (see 11.1)	Temporary wider rule, then restricted again

12. What I Learned

- VPC peering needs routes in both directions, and a peering connection is not transitive.
- Routes and security groups both have to allow traffic. Removing one ICMP rule broke the ping.
- A bastion host lets private servers stay without public IPs while remaining reachable for administration.
- Security group sources should be as narrow as possible, such as /32 addresses. The private key must stay on the laptop.
- Cleaning up resources after testing avoids unnecessary cost.

13. Cleanup Evidence

After saving all evidence, I deleted every resource created for the lab in the order below so nothing continues to cost money.

Order	Action	Done
1	Terminate ewb-bastion, ewb-app-a and ewb-app-b, and wait for the state Terminated	☑
2	Delete the peering routes from the three route tables	☑
3	Delete the peering connection ewb-a-b-peer	☑
4	Delete the security groups ewb-sg-bastion, ewb-sg-app-a and ewb-sg-app-b	☑
5	Delete the subnets ewb-a-public, ewb-a-private and ewb-b-private	☑
6	Delete the route tables ewb-rt-a-public, ewb-rt-a-private and ewb-rt-b-private	☑
7	Detach and delete the internet gateway ewb-igw-a	☑
8	Delete the VPCs ewb-vpc-a and ewb-vpc-b	☑
9	Delete the key pair ewb-lab and the local .pem file	☑
10	Check the Project tag filter and Billing show nothing left	☑

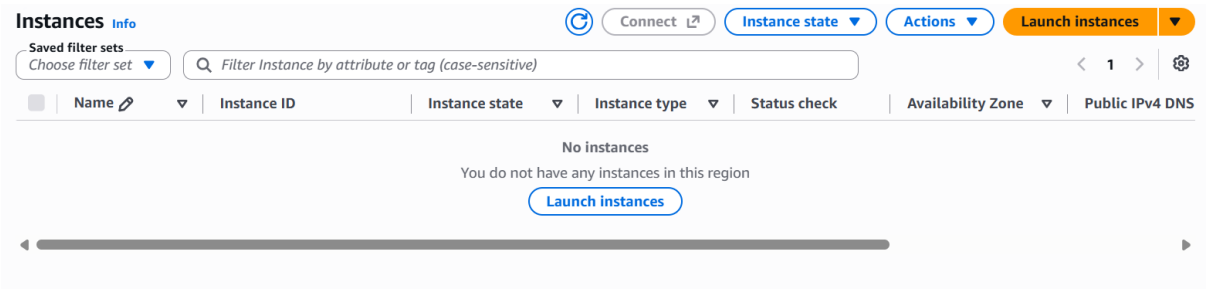


Figure 30: Instances terminated

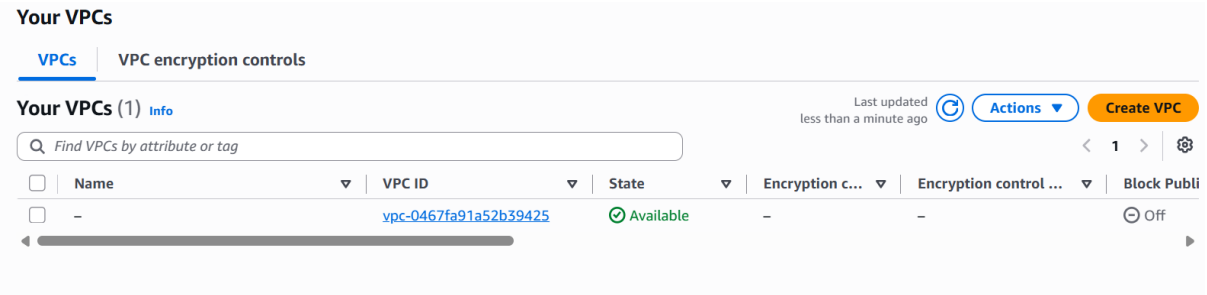


Figure 31: Lab VPCs deleted; only the default VPC remains